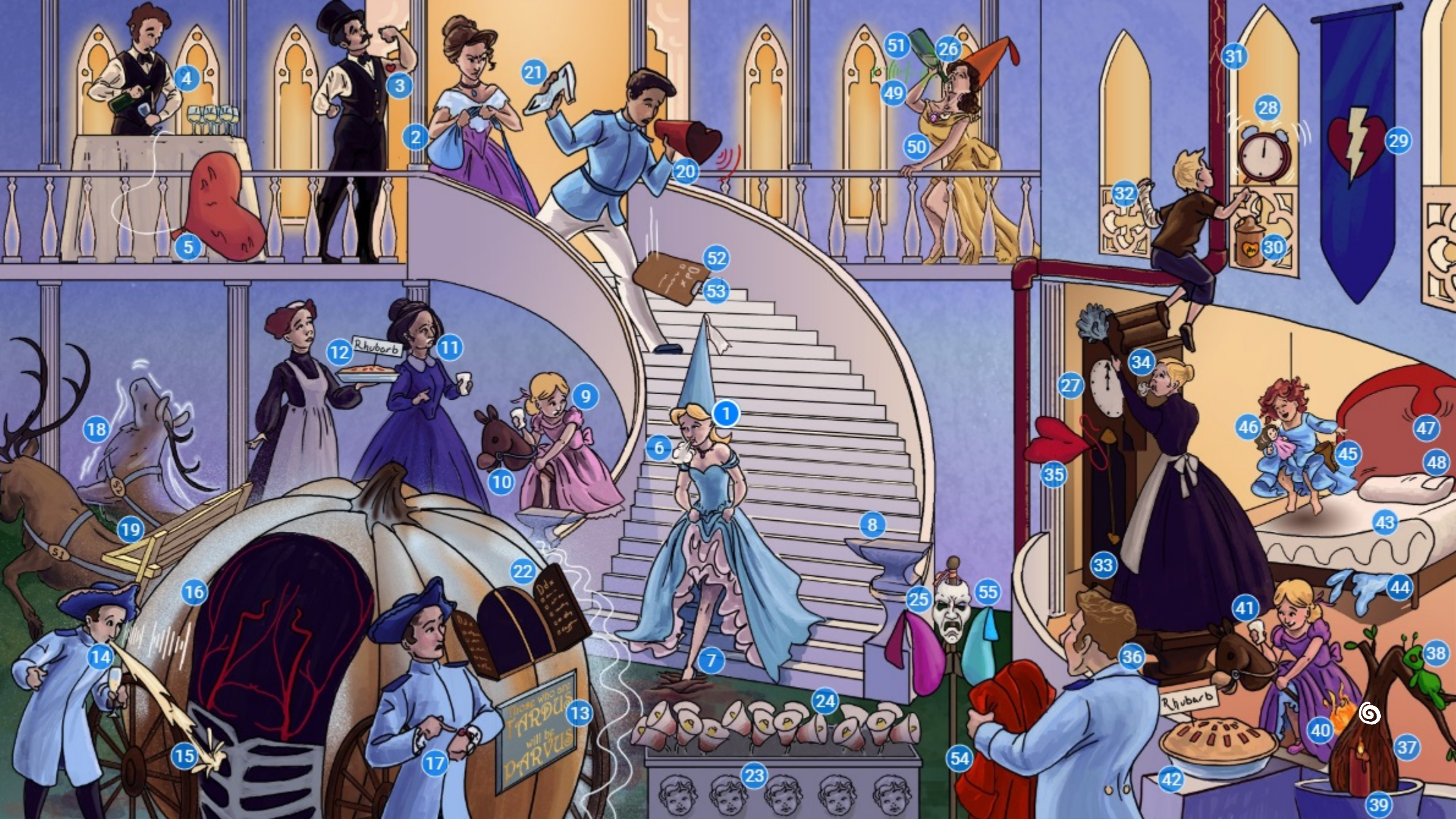




3.1 - Aortic Stenosis & Aortic Regurgitation SOAP

1) Cone-shaped hat on princess:

Aortic valve

2) Sister squeezing cone-shaped hat:

Aortic stenosis = most common cause of left ventricular outflow obstruction

3) Enlarged left muscle:

LV outflow obstruction results in **LV hypertrophy**

4) Struggling to fill glasses:

Aortic stenosis → Chronic LVH → **DIASTOLIC dysfunction** → HEART FAILURE

5) Floppy heart balloon:

Aortic stenosis → Chronic LVH → DIASTOLIC dysfunction → **HEART FAILURE**

6) Princess blowing puff of air:

Dyspnea on exertion = most common symptom of **AS (Chief Complaint)**

7) Foot sinking into mud:

SYNCOPE with exertion is a common symptom of **AS (Chief Complaint)**

8) Anvil bannister:

ANGINA is a common symptom of severe **AS (HPI)**

9) Young child:

CALCIFICATION of congenital BICUSPID aortic valve = most common cause of **AS** presenting **before age 60 (Demographics)**

10) Toy horse with bicuspid mouth:

CALCIFICATION of congenital BICUSPID aortic valve = most common cause of AS presenting before age 60 (Demographics)

11) Old stepmother with milk mustache:

Age-related sclerocalcific change of normal tricuspid aortic valve = most common cause of AS presenting after age 60 (**Demographics**)

12) Rhubarb pie:

Rheumatic heart disease = most common cause of **AS** worldwide (**PMHx**)

13) “Those who are TARDUS will be PARVUS”:

Pulsus parvus et tardus = weak & slow-rising carotid pulse classic in **AS** (**Physical Exam**)

14) Systolic spray with murmur lines:

Classic **AS** murmur = “**crescendo-decrescendo**” **SYSTOLIC** ejection murmur best heard at 2nd right intercostal space, radiating to the CAROTIDS (**Cardiac Auscultation**)

15) Spray hitting 2nd right intercostal space of carriage:

Classic **AS** murmur = “crescendo-decrescendo” **SYSTOLIC** ejection murmur **best heard at 2nd right intercostal space**, radiating to the CAROTIDS (**Cardiac Auscultation**)

16) Carotid carriage next to spray:

Classic **AS** murmur = “crescendo-decrescendo” **SYSTOLIC** ejection murmur best heard at 2nd right intercostal space, **radiating to the CAROTIDS** (**Cardiac Auscultation**)

17) Angry attendant pointing at watch:

Timing of murmur correlates with severity of **AS** (i.e., LATE peaking murmur = SEVERE AS) (**Cardiac Auscultation**)

18) Disappearing S2 deer with single antler:

SOFT, SINGLE S2 is suggestive of **AS** (**Cardiac Auscultation**)

19) Stiff 4-shaped chair:

S4 gallop can be heard if ventricular hypertrophy develops (**Cardiac Auscultation**)

20) ECHO bullhorn:

Transthoracic echocardiogram = gold standard for diagnosis of **AS** (**Echo**)

21) Glass “<1” slipper:

Severe **AS** = **aortic valve area <1 cm²** on echocardiogram (**Echo**)

22) **Aortic Stenosis** DDx Clipboard:

For further workup of Aortic Stenosis, see Chronic Dyspnea DDx, Chest Pain DDx

23) Smiling cherubs on planter:

ASYMPTOMATIC patients with mild to moderate **AS** should receive **SERIAL ECHOS** to monitor disease progression **(1)**

24) Row of bullhorn flowers:

ASYMPTOMATIC patients with mild to moderate **AS** should receive **SERIAL ECHOS** to monitor disease progression **(1)**

25) Cone-shaped hat on sad coat rack:

Most patients with severe **AS** or **SYMPTOMATIC AS** should undergo **AV replacement** surgery **(2)**

26) Champagne regurgiting from sister with conical hat:

Aortic REGURGITATION is caused by inadequate closure of aortic valve leaflets

27) Chronically old grandfather clock:

CHRONIC AR has a long asymptomatic compensated phase before symptoms develop

28) Acutely ringing alarm clock:

ACUTE AR evolves over hours to days → usually a **MEDICAL EMERGENCY** due to inability of LV to compensate for increased volume

29) Heart lightning bolt:

ACUTE AR most commonly presents with **CARDIOGENIC SHOCK (Chief Complaint)**

30) Flame inside heart lantern:

ENDOCARDITIS = most common cause of **ACUTE AR (HPI)**

31) Red pipe with longitudinal crack:

AORTIC DISSECTION = 2nd most common cause of **ACUTE AR (HPI)**

32) Arm cast:

Trauma (accidental or iatrogenic) to the aortic valve can result in **ACUTE AR (HPI)**

33) Dilated dress:

Chronic **AR** → chronically increased LVEDV → eccentric hypertrophy → **LV dilation**

34) Maid blowing puff of air:

: Severe **AR** commonly presents with **DYSPNEA on exertion** & other signs of **HEART FAILURE (Chief Complaint)**

35) Floppy heart balloon next to grandfather clock:

Severe **AR** commonly presents with **DYSPNEA on exertion** & other signs of **HEART FAILURE (Chief Complaint)**

36) Angina anvil step stool:

Severe **AR** may present with **ANGINA** due to ↓ diastolic pressure → ↓ coronary perfusion (**Chief Complaint**)

37) Potted aorta tree with wide trunk:

In developed countries, chronic **AR** is most commonly caused by **aortic root dilation** (e.g., 2/2 MARFAN syndrome, Ehlers-Danlos syndrome, osteogenesis imperfecta, large-vessel VASCULITIS, syphilis AORTITIS) (**PMHx**)

38) Martian doll:

In developed countries, chronic **AR** is most commonly caused by aortic root dilation (e.g., 2/2 **MARFAN syndrome**, Ehlers-Danlos syndrome, osteogenesis imperfecta, large-vessel VASCULITIS, syphilis AORTITIS) (**PMHx**)

39) Flaming candle by tree:

In developed countries, chronic **AR** is most commonly caused by aortic root dilation (e.g., MARFAN syndrome, Ehlers-Danlos syndrome, osteogenesis imperfecta, **large-vessel VASCULITIS**, syphilis AORTITIS) (**PMHx**)

40) Bark of aorta tree on fire:

In developed countries, chronic **AR** is most commonly caused by aortic root dilation (e.g., MARFAN syndrome, Ehlers-Danlos syndrome, osteogenesis imperfecta, large-vessel VASCULITIS, **syphilis AORTITIS**) (**PMHx**)

41) Little girl with bicuspid horse:

Congenital bicuspid aortic valve is a common cause of chronic **AR** → presents at a younger age than AS 2/2 congenital bicuspid aortic valve (**PMHx**)

42) Rhubarb pie on sill:

Rheumatic heart disease = most common cause of chronic **AR** in developing countries (**PMHx**)

43) Widened peak and trough:

Chronic **AR** causes **widened pulse pressure** → large difference (>100 mmHg) between systolic & diastolic BPs (**Vitals**)

44) Water pulsing from bed:

Corrigan pulse = bounding, “**water hammer**” pulse in chronic **AR** best felt in upper extremities & carotids (**Physical Exam**)

45) Water gun holster:

Traube’s sign = “**pistol shot**” sounds heard over the femoral arteries in chronic **AR** (**Physical Exam**)

46) Dolls head bobbing:

de Musset’s sign = **head bobbing** with each heartbeat seen in chronic **AR** (**Physical Exam**)

47) Shaking uvula-shaped headboard:

Müller’s sign = **visible pulsations of uvula** with each heartbeat seen in chronic **AR** (**Physical Exam**)

48) Shaking finger-shaped pillow:

Quincke's pulses = alternating flushing and blanching of nail beds seen in chronic **AR** (Physical Exam)

49) Murmur lines from sister wearing diamonds:

Murmur of chronic **AR** = **DIASTOLIC decrescendo** murmur best heard at left sternal border (**Cardiac Auscultation**)

50) Sister gripping railing and squatting:

Murmur of chronic **AR** **increases with handgrip/squatting** & decreases with Valsalva (**Cardiac Auscultation**)

51) Half-full champagne bottle:

Regurgitant fraction >50% on echo is one definition of severe **AR** (Echo)

52) Chronic Aortic Regurgitation DDx Clipboard:

For further workup of Chronic **Aortic Regurgitation**, see Chronic Dyspnea DDx and Chest Pain DDx

53) Acute **Aortic Regurgitation** DDx Clipboard:

For further workup of Acute Aortic Regurgitation, see Cardiogenic & Obstructive Shock SOAP

54) Dilated red sleeves:

Vasodilating drugs (e.g., ACE inhibitors, nifedipine, hydralazine) can be used in patients with severe **AR** by ↓ afterload & ↑ forward flow(1)

55) Second replacement cone hat on sad coat rack:

Aortic valve replacement is indicated for all SYMPTOMATIC patients with severe **AR** (2)